7.7.1 Expansion of the Operator $\hat{\mathcal{L}}$

Consider a Langevin splitting method using the A, B and O splitting pieces, where we compose exact solves of each of the vector fields in a prescribed sequence to define our discretization scheme. Because we solve each piece exactly, we are able to write down the action of each individual solve on the initial distribution in terms of the vector field's evolution operator (using the respective generators for each piece as in [99]), where

$$\begin{aligned}
\left(e^{t\mathcal{L}_{\mathrm{A}}}\phi\right)(\boldsymbol{q},\boldsymbol{p}) &= \phi(\boldsymbol{q}+t\boldsymbol{M}^{-1}\boldsymbol{p},\boldsymbol{p}),\\
\left(e^{t\mathcal{L}_{\mathrm{B}}}\phi\right)(\boldsymbol{q},\boldsymbol{p}) &= \phi(\boldsymbol{q},\boldsymbol{p}-t\nabla U(\boldsymbol{q})),\\
\left(e^{t\mathcal{L}_{\mathrm{O}}}\phi\right)(\boldsymbol{q},\boldsymbol{p}) &= \int_{\mathcal{P}} \phi\left(\boldsymbol{q}, e^{-\gamma t}\boldsymbol{p}+\zeta\boldsymbol{M}^{1/2}\boldsymbol{x}\right)\frac{e^{-|\boldsymbol{x}|^2/2}}{(2\pi)^{\mathrm{N_c}/2}}\mathrm{d}\boldsymbol{x},
\end{aligned}$$

where $\zeta = \left[\mathrm{k_B T}\left(1-e^{-2\gamma t}\right)\right]^{1/2}$. The overall generator for an associated Langevin splitting method $\hat{\mathcal{L}}$ can be computed by composing each individual operator associated to the terms in the splitting. For example, for the method labelled by the string $[\![\mathrm{ABAO}]\!]$, we have

$$\exp(h\hat{\mathcal{L}}_{[\![\mathrm{ABAO}]\!]}) = \exp((h/2)\mathcal{L}_{\mathrm{A}})\exp(h\mathcal{L}_{\mathrm{B}})\exp((h/2)\mathcal{L}_{\mathrm{A}})\exp(h\mathcal{L}_{\mathrm{O}}),$$

where importantly note the apparent reversal of the sequence of products, known as the *vertauschungssatz* (see the discussion in [164, Section III.5.1]), which is a consequence of the relation

$$\phi(\mathscr{U}_t^{\mathrm{f}_1}(z)) = \left(e^{t\mathcal{L}_{\mathrm{f}_1}}\phi\right)(z),$$

for some vector field f_1 with associated flow map $\mathscr{U}_t^{\mathrm{f}_1}$ and generator $\mathcal{L}_{\mathrm{f}_1}$, where the choice of $\phi(z) = \mathscr{U}_t^{\mathrm{f}_2}(z) = (e^{t\mathcal{L}_{\mathrm{f}_2}}\mathrm{Id})(z)$ demonstrates this reversal.

We will find it more convenient to work with the adjoint of the perturbed generator, denoted $\hat{\mathcal{L}}^\dagger$, which we may compute as

$$\exp(h\hat{\mathcal{L}}^\dagger_{[\![\mathrm{ABAO}]\!]}) = \exp(h\mathcal{L}^\dagger_{\mathrm{O}})\exp((h/2)\mathcal{L}^\dagger_{\mathrm{A}})\exp(h\mathcal{L}^\dagger_{\mathrm{B}})\exp((h/2)\mathcal{L}^\dagger_{\mathrm{A}}).$$

Under the propagation of the numerical scheme (or the discretized dynamics), the Fokker-Planck equation gives us that the evolution of a density of points ρ is given by

$$\frac{\partial}{\partial t}\rho(z,t) = \hat{\mathcal{L}}^\dagger\rho,$$

with the important case of its invariant distribution satisfying the partial differential equation

$$\hat{\mathcal{L}}^\dagger \hat{\rho} = 0.$$

Our goal in the following section is to find $\hat{\mathcal{L}}^\dagger$ and solve the PDE for the distribution $\hat{\rho}$ by using our ansatz (7.17). The solution gives us the corrupted invariant distribution sampled by our numerical scheme, allowing us to correct our observed averages to achieve more accurate results. It also gives us a way of evaluating which schemes are best for sampling, by evaluating the error introduced by the discretization.

The terms of the perturbation expansion for $\hat{\mathcal{L}}^\dagger$ can be computed using the Baker-Campbell-Hausdorff (BCH) expansion already introduced in Chap. 3. Recall that for linear operators X and Y, we can write the composition of their exponentials as

$$\exp(hX)\exp(hY) = \exp(hZ),$$

where we have the expansion

$$Z = X + Y + \frac{h}{2}[X, Y] + \frac{h^2}{12}([X, [X, Y]] + [Y, [Y, X]]) + \mathcal{O}(h^3).$$

The notation $[X, Y] \stackrel{\text{def}}{=} XY - YX$ represents the commutator of the operators X and Y. By repeated application of the BCH formula, we can obtain such an expansion for any splitting-based method, however the terms grow in complexity at successive powers of h, requiring commutators of increasing length that make computation cumbersome. Additionally any careful mathematician should question the convergence of this sequence when computing $\hat{\mathcal{L}}$, as we deal only with formal series that offer no guidance on the boundedness of the resulting terms, which will involve high order derivatives in both $\boldsymbol{q}$ and $\boldsymbol{p}$.

To offer some reassurance, let us define the evolution operator for our numerical method as

$$\mathcal{P}_t = e^{t\hat{\mathcal{L}}} = e^{t\mathcal{L}_1} e^{t\mathcal{L}_2} \dots e^{t\mathcal{L}_k},$$

where the numerical method has k-many stages with each $\mathcal{L}_i$ proportional to one of $\mathcal{L}_\mathrm{A}$, $\mathcal{L}_\mathrm{B}$ and $\mathcal{L}_\mathrm{O}$. We can make use of Taylor's theorem to write the identity

$$\mathcal{P}_h = \mathcal{P}_0 + h \left.\frac{\mathrm{d}\mathcal{P}_t}{\mathrm{d}t}\right|_{t=0} + \dots + \frac{h^n}{n!} \left.\frac{\mathrm{d}^n\mathcal{P}_t}{\mathrm{d}t^n}\right|_{t=0} + \frac{h^{n+1}}{n!} \int_0^1 (1-\lambda)^n \left.\frac{\mathrm{d}^{n+1}\mathcal{P}_t}{\mathrm{d}t^{n+1}}\right|_{t=\lambda h} \mathrm{d}\lambda,$$